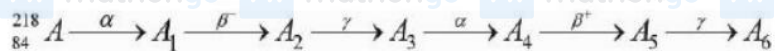


Q1 - 24 January - Shift 1

Consider the following radioactive decay process



The mass number and the atomic number A_6 are given by :

- (1) 210 and 82 (2) 210 and 84
(3) 210 and 80 (4) 211 and 80

Space for your notes:

Q2 - 24 January - Shift 2

The energy released per fission of nucleus of ${}^{240}\text{X}$ is 200 MeV. The energy released if all the atoms in

120g of pure ${}^{240}\text{X}$ undergo fission is _____ $\times 10^{25}$

MeV.

(Given $N_A = 6 \times 10^{23}$)

Space for your notes:

Q3 - 25 January - Shift 1

The ratio of the density of oxygen nucleus (${}^{16}_8\text{O}$)

and helium nucleus (${}^4_2\text{He}$) is

- (1) 4:1 (2) 8:1
(3) 1:1 (4) 2:1

Space for your notes:

Q4 - 29 January - Shift 1

If a radioactive element having half-life of 30 min is undergoing beta decay, the fraction of radioactive element remains undecayed after

90 min. will be :

- (1) $\frac{1}{8}$ (2) $\frac{1}{16}$
(3) $\frac{1}{4}$ (4) $\frac{1}{2}$

Space for your notes:

Q5 - 29 January - Shift 1

A radioactive element ${}_{92}^{242}\text{X}$ emits two α -particles, one electron and two positrons. The product nucleus is represented by ${}_{\text{P}}^{234}\text{Y}$. The value of P is

Space for your notes:

Q6 - 29 January - Shift 2

Substance A has atomic mass number 16 and half life of 1 day. Another substance B has atomic mass

number 32 and half life of $\frac{1}{2}$ day. If both A and B

simultaneously start undergo radio activity at the same time with initial mass 320 g each, how many total atoms of A and B combined would be left after 2 days.

- (1) 3.38×10^{24} (2) 6.76×10^{24}
(3) 6.76×10^{23} (4) 1.69×10^{24}

Space for your notes:

Q7 - 30 January - Shift 2

Given below are two statements : one is labelled as **Assertion A** and the other is labelled as **Reason R**.

Assertion A : The nuclear density of nuclides ${}^{10}_5\text{B}$, ${}^6_3\text{Li}$, ${}^{56}_{26}\text{Fe}$, ${}^{20}_{10}\text{Ne}$ and ${}^{209}_{83}\text{Bi}$ can be arranged as $\rho_{\text{Bi}}^{\text{N}} > \rho_{\text{Fe}}^{\text{N}} > \rho_{\text{Ne}}^{\text{N}} > \rho_{\text{B}}^{\text{N}} > \rho_{\text{Li}}^{\text{N}}$.

Reason R : The radius R of nucleus is related to its mass number A as $R = R_0 A^{1/3}$, where R_0 is a constant.

In the light of the above statement, choose the **correct** answer from the options given below :

- (1) Both **A** and **R** are true and **R** is the correct explanation of **A**
- (2) **A** is false but **R** is true
- (3) **A** is true but **R** is false
- (4) Both **A** and **R** are true but **R** is **NOT** the correct explanation of **A**

Space for your notes:

Q8 - 30 January - Shift 2

A radioactive nucleus decays by two different process. The half life of the first process is 5 minutes and that of the second process is 30s.

The effective half-life of the nucleus is calculated

to be $\frac{\alpha}{11}$ s. The value of α is _____.

Space for your notes:

Q9 - 31 January - Shift 1

A free neutron decays into a proton but a free proton does not decay into neutron. This is because

- (1) neutron is an uncharged particle
- (2) proton is a charged particle
- (3) neutron is a composite particle made of a proton and an electron
- (4) neutron has larger rest mass than proton

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)

Q2 (6)

Q3 (3)

Q4 (1)

Q5 (87)

Q6 (1)

Q7 (2)

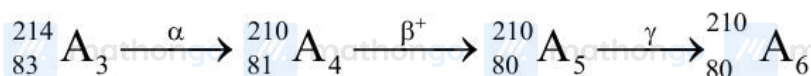
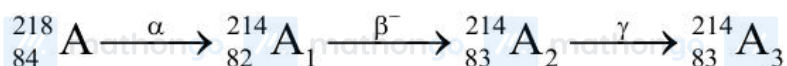
Q8 (300)

Q9 (4)

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Q1 (3)



Q2 (6)

$$\text{No. of mole} = \frac{120}{240} = \frac{1}{2}$$

$$\text{No. of molecules} = \frac{1}{2} \times N_A$$

$$\text{Energy released} = \frac{1}{2} \times 6 \times 10^{23} \times 200$$

$$= 6 \times 10^{25} \text{ MeV}$$

Q3 (3)

Nuclear density is independent of mass number

$$\text{As nuclear density} = \frac{\text{Au}}{\frac{4}{3}\pi R^3}$$

$$\text{Also, } R = R_0 A^{\frac{1}{3}}$$

$$\text{And } R^3 = R_0^3 A$$

$$\Rightarrow \text{Nuclear density} = \frac{\text{Au}}{\frac{4}{3}\pi R_0^3 A}$$

$$\text{Nuclear density} = \frac{3u}{4\pi R_0^3}$$

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Q4 (1)

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^{t/t_{1/2}} = \left(\frac{1}{2}\right)^{\frac{90}{30}}$$

$$\frac{N}{N_0} = \left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

Q5 (87)

$$P = 92 - 2 - 2 + 1 - 1 - 1$$

$$P = 92 - 5$$

$$P = 87$$

Q6 (1)

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$$(N_0)A = \frac{320}{16} = 20 \text{ moles}$$

$$(N_0)B = \frac{320}{32} = 10 \text{ moles}$$

$$N_A = \frac{(N_0)_A}{(2)^{2/1}} = \frac{20}{4} = 5$$

$$N_B = \frac{(N_0)_B}{(2)^{2/.5}} = \frac{10}{2^4} = 0.625$$

$$\text{Total } N = 5.625$$

$$\begin{aligned} \text{No. of atoms} &= 5.625 \times 6.023 \times 10^{23} \\ &= 3.38 \times 10^{24} \end{aligned}$$

Ans. (1)

Q7 (2)

Nuclear density is independent of A

Q8 (300)

$$\frac{dN_1}{dt} = -\lambda_1 N \quad \frac{dN_2}{dt} = -\lambda_2 N$$

$$\frac{dN}{dt} = -(\lambda_1 + \lambda_2) N$$

$$\Rightarrow \lambda_{eq} = \lambda_1 + \lambda_2$$

$$\Rightarrow \frac{1}{t_{1/2}} = \frac{1}{300} + \frac{1}{30} = \frac{11}{300}$$

$$\Rightarrow t_{1/2} = \frac{300}{11}$$

Q9 (4)

As neutron has more rest mass than proton it will require energy to decay proton into neutron.

Option 4.